

SELENE: Selective Empathy LoRA with Factor Gates

for English–Korean Affective Transfer

Jaehong Kim, Seohyeon Yoo, and Jaehyuk Cho

Abstract—Large language models that perform well on English empathetic dialogue often fail to transfer that competence to Korean, where empathy is shaped by relational stance, honorifics, and high-context support strategies. We propose SELENE (Selective Empathy LoRA with EN–KO factor gatEs): a parameter-efficient EN→KO affective transfer system on a frozen Qwen/Qwen3.5–9B backbone. SELENE (i) pretrains an English empathy LoRA with an affect head on EmpatheticDialogues, (ii) assigns factor-wise share/relearn/suppress gates from EN–KO divergence and frozen probes, and (iii) adapts a Korean LoRA with multitask affect, strategy, and relation supervision on AI Hub dialogues. Unlike language×task adapter merging alone, SELENE uses factor-wise gates to *diagnose* which empathy axes (Affect / Strategy / Relation) should be shared versus relearned, then realizes a mixed gate pattern with a structural Factor-Bank (EN-merge + KO LoRA + two-pass affect) and gate-conditioned losses/LRs—not per-factor LoRA parameters. We evaluate two directions: Direction I Korean A/S/R accuracy versus non-selective baselines, and Direction II return-to-English performance after KO adaptation (gain versus forgetting), with KoED held out for cultural evaluation. On Qwen3.5-9B, Stage 2 assigns Affect=share and Strategy/Relation=relearn. Factor-LoRA Bank (F16/F32) raises Direction I Avg Acc by ≈ 4.5 pp over matched Blind/Scratch baselines on full AI Hub valid ($n=3182$), while KO-scratch collapses Direction II (-34.4 pp) but Bank retains English emotion Acc within 1.3 pp. Multi-seed and component ablations confirm that EN-merge + two-pass affect—not soft scalar gating alone—drives the gain.

Index Terms—Affective computing, empathetic dialogue, cross-lingual transfer, LoRA, cultural adaptation, Korean NLP, support strategies.

I. INTRODUCTION

EMPATHETIC response generation is a core affective computing problem that couples emotion understanding with socially appropriate generation [1], [2], [13]. Open-domain empathy benchmarks such as EmpatheticDialogues (ED) [1] and emotional-support corpora such as ESC [2] have driven rapid progress in English, and large language models (LLMs) further improve fluency on affective tasks [4], [12]. Yet multilingual competence does not imply multicultural empathy [3], [17], [18]: models that score well in English frequently violate Korean relational norms, over-empathize,

or select support strategies that are pragmatically odd in honorific, high-context settings [3], [11].

A common remedy is full fine-tuning on Korean dialogues. This is compute-heavy and risks catastrophic forgetting of English affective skills [9], [22]. Parameter-efficient fine-tuning (PEFT), especially LoRA [5] and modular adapters [6], [21], reduces cost and keeps the backbone frozen. However, typical EN→KO transfer still treats empathy as a *monolithic* skill: either copy an English adapter blindly, train Korean from scratch, or merge language×task adapters without asking which *empathy factors* should be shared versus relearned [7], [8].

Psychological and dialogue research argues that empathy is multidimensional [10]: affective sharing, perspective-taking, helping strategies, and culturally regulated display rules are separable. ESC further shows that support *strategy*—not emotion mirroring alone—drives perceived helpfulness [2], [13]. KoED demonstrates a measurable cultural empathy gap between ED and culturally reconstructed Korean dialogues [3]. Together, these observations motivate *factor-selective* transfer rather than uniform adapter reuse.

This paper asks: can we transfer English empathy to Korean selectively at the level of theoretically motivated affective factors, while keeping the backbone frozen and preserving return-to-English performance?

Our contributions are:

- 1) A theory-aligned A/C/S/R factor schema for empathetic dialogue (Affect, Cognition, Strategy, Relation), anchored in empathy theory, ESC strategies, and Korean relational norms [2], [3], [10], [11].
- 2) **SELENE** / Factor-Bank: English LoRA pretraining, factor-wise share/relearn/suppress *diagnosis*, and a Korean adaptation that implements mixed gates via EN-merge + fresh KO LoRA, two-pass affect, and gate-conditioned head losses/LRs on frozen Qwen/Qwen3.5–9B.
- 3) A dual evaluation protocol—Direction I (EN→KO affective accuracy) and Direction II (return-to-EN after KO)—plus stage-wise ablations that isolate EN pretraining, gates, and multitask heads.
- 4) A unified EN–KO data pipeline over EmpatheticDialogues, KoED, KorEmpatheticDialogues, and AI Hub under one JSONL schema, with factor-wise EN–KO comparison figures that justify selective gates.

J. Kim, S. Yoo, and J. Cho are with the Department of Software Engineering, Jeonbuk National University, Jeonju, Republic of Korea (corresponding author: J. Cho; e-mail: chojh@jbnu.ac.kr).

Manuscript received XXX XX, 2026; revised XXX XX, 2026. Working draft targeting *IEEE Transactions on Affective Computing*. Backbone: Qwen/Qwen3.5–9B.

II. RELATED WORK

A. Empathetic and Emotional Support Dialogue

EmpatheticDialogues remains the standard open-domain empathy benchmark for emotion-grounded response generation [1]. Subsequent systems emphasize mixture-of-experts emotion routing [14], emotion mimicking [15], and commonsense-aware empathy [16]. Emotional Support Conversation (ESC) shows that staged helping strategies (exploration→comforting→action) improve support effectiveness beyond affect mirroring [2]. Computational empathy metrics for support text further highlight explorations, interpretations, and emotional reactions as complementary axes [13]. EmoBench evaluates multi-faceted emotional intelligence in LLMs [4], and broad TAFFC evaluations report uneven LLM behavior across affective tasks [12]. Our Strategy (S) axis follows ESC and is operationalized with AI Hub labels (advice / encourage / comfort / agree), while Affect (A) uses emotion labels from ED and AI Hub.

B. Cultural Empathy and Multilingual Evaluation

KoED is our problem mother paper [3]: multilingual LLMs underperform on culturally reconstructed Korean dialogues relative to ED (emotion Acc 36.84 → 32.69 on shared emotions), and cultural appropriateness favors Korean-aligned models. This aligns with broader evidence that LLMs exhibit Western-centric cultural defaults even when prompted in other languages [17], and that multilingual benchmarks must separate linguistic fluency from cultural competence [18], [19]. Cross-cultural psychology motivates Relation (R) through display rules and emotion regulation under social context [11]. SELENE does not replace culturally aligned pretraining; instead it provides a PEFT mechanism to *selectively* adapt English empathy factors toward Korean A/S/R supervision while freezing a shared backbone.

C. Parameter-Efficient Cross-Lingual Transfer

Adapter modules [21] and LoRA [5] enable low-rank task adaptation without updating all backbone weights. MAD-X separates language and task adapters for modular cross-lingual transfer [6]. AdaMergeX merges task/language LoRAs with adaptive arithmetic and reports large gains over MAD-X-style baselines [7]. FLARE fuses EN/target representations inside LoRA bottlenecks and improves target-language accuracy over plain LoRA [8]. Strong multilingual encoders such as XLM-R underpin much of this transfer stack [20]. On the forgetting side, classical continual-learning results warn that sequential adaptation can erase prior skills [22], and LF-MLF shows that reducing source-side forgetting improves retained multilingual transfer [9]—motivating our Direction II return-to-EN protocol. These methods operate at language×task granularity; SELENE instead *diagnoses* empathy factors A/S/R with share/relearn/suppress gates and realizes mixed gates via Factor-Bank routing plus gate-conditioned optimization.

III. PROBLEM FORMULATION

Let x be a dialogue context and y an empathetic listener response. We do *not* claim a unique latent decomposition of empathy. Instead we give operational definitions, testable propositions, and analytic consequences of the gate map that jointly justify factor-selective transfer.

A. Definitions

Definition 1 (Theory-aligned factors). Let $\mathcal{K} = \{A, C, S, R\}$. For each factor $k \in \mathcal{K}$, an observable label space $\mathcal{Y}_k$ is fixed a priori by theory and corpus annotation (Table 1): A = emotion, C = situation/appraisal proxy, S = support strategy, R = relation type [1]–[3], [10], [11]. When y_C is unavailable as a discrete label, C is treated as a proxy factor and excluded from gated decisions unless otherwise stated.

Definition 2 (Factor representation and head). Let $h_\theta(x) \in \mathbb{R}^d$ be the pooled hidden state of a frozen backbone under a chosen adapter state. In this paper $P_k=I$ (shared encoder; separate heads): $\hat{y}_k = f_k(h_\theta(x))$. When Factor-Bank uses two-pass affect, the affect head reads a share-path encoding h^{share} while strategy/relation heads read a relearn-path encoding h^{KO} (Definition 5).

Definition 3 (Transfer score and gate map). On Korean labeled pairs $\{(x_i^{\text{KO}}, y_{k,i})\}$, the transfer score is

$$s_k = \text{Acc}(f_k^{\text{probe}}; h_\theta(x^{\text{KO}}), y_k^{\text{KO}}) \in [0, 1], \quad (1)$$

optionally mixed with an EN–KO alignment term. Fix thresholds $1 \geq \tau_{\text{share}} > \tau_{\text{suppress}} \geq 0$. The three intervals $[0, \tau_{\text{suppress}}]$, $(\tau_{\text{suppress}}, \tau_{\text{share}})$, $[\tau_{\text{share}}, 1]$ form a partition of $[0, 1]$, so there is a unique map

$$g_k = \Gamma(s_k) = \begin{cases} \text{share}, & s_k \geq \tau_{\text{share}}, \\ \text{suppress}, & s_k \leq \tau_{\text{suppress}}, \\ \text{relearn}, & \text{otherwise}, \end{cases} \quad (2)$$

sending each score to one gate. Equip gates with the total order $\text{suppress} \prec \text{relearn} \prec \text{share}$ (“more sharing” is larger).

Definition 4 (Single-bank init operators). Let M_{EN} be the Stage 1 English LoRA and Rand a matched-rank random init. Legacy single-bank modes use one shared adapter M_{KO} with a global seed

$$\mathcal{I}_{\text{global}}(g) = \begin{cases} M_{\text{EN}}, & g = \text{share}, \\ \text{Rand}, & g = \text{relearn}, \\ \rho M_{\text{EN}}, & g = \text{suppress}, \end{cases} \quad (3)$$

($\rho \in [0, 1]$). Soft-share always takes M_{EN} ; auto uses majority vote over $\{g_k\}$. These modes do not allocate a separate LoRA per factor.

Definition 5 (Factor-Bank realization of mixed gates). When Stage 2 yields the mixed pattern $g_A=\text{share}$ and

$g_S, g_R = \text{relearn}$ (Proposition 3), Factor-Bank implements it with one trainable KO LoRA plus a share path—not $\{M_k\}$:

$$\theta^{\leftarrow} \leftarrow \text{Merge}(\theta, M_{\text{EN}}), \quad (4)$$

$$M_{\text{KO}} \leftarrow \text{Rand}, \quad (5)$$

$$h^{\text{share}}(x) = h_{\theta^{\leftarrow}}(x) \quad (\text{adapters off}), \quad (6)$$

$$h^{\text{KO}}(x) = h_{\theta^{\leftarrow} + M_{\text{KO}}}(x). \quad (7)$$

Affect uses $f_A(h^{\text{share}})$; LM / Strategy / Relation use h^{KO} . Factor gates further scale head losses and learning rates

$$\lambda_k \leftarrow \lambda_k \cdot \sigma(g_k), \quad \eta_k \leftarrow \eta_{\text{head}} \cdot \varsigma(g_k), \quad (8)$$

with $\sigma(\text{share}) < 1 < \sigma(\text{relearn})$ (and likewise for ς), plus an optional early curriculum that amplifies relearn heads. Thus g_k acts on **forward routing** (A vs. S/R path), **loss weights**, and **head LR**s—not on separate LoRA tensors per factor.

English pretraining yields a transferable affect direction; Korean requires culture-specific realization especially for z_S and z_R [3]. The learning objective maximizes Korean A/S/R accuracy (Direction I) subject to limited degradation of English emotion recognition after KO adaptation (Direction II).

B. Propositions and Gate Lemmas

Proposition 1 (Separability). For each $k \in \{A, S, R\}$, a linear probe f_k on frozen $h_{\theta}(x^{\text{KO}})$ satisfies $\text{Acc}_k \geq \text{chance}_k + \delta$ for a pre-registered margin $\delta > 0$ (e.g., $\delta=0.1$), under a held-out KO split.

Proposition 2 (Partial non-redundancy). For $k \neq k'$, cross-factor probe accuracy $\text{Acc}(f_k \rightarrow y_{k'})$ is significantly lower than $\text{Acc}(f_k \rightarrow y_k)$, and/or Cramér’s $V(y_k, y_{k'})$ remains below a redundancy ceiling V_0 .

Proposition 3 (Transfer ordering). Under an EN Stage 1 encoder, transfer scores obey $s_A \geq \tau_{\text{share}}$ and $s_S, s_R < \tau_{\text{share}}$ (equivalently $s_A, s_C \gg s_S, s_R$ when C is scored), supporting share for affect(/cognition) and relearn for strategy/relation.

Proposition 4 (Gated intervention). With matched backbone, rank, and budget, $\text{DirI}(\text{SELECT}) > \text{DirI}(\text{Blind})$ and $|\Delta_{\text{DirII}}(\text{SELECT})| \leq |\Delta_{\text{DirII}}(\text{Blind})|$, where Blind uses $\Gamma \equiv \text{share}$.

Lemma 1 (Monotone gate and local stability). Under Definition 3, Γ is (i) monotone nondecreasing w.r.t. $\prec$, and (ii) locally constant away from thresholds. Precisely: if $s \leq s'$, then $\Gamma(s) \preceq \Gamma(s')$. Moreover, if $\text{dist}(s, \{\tau_{\text{suppress}}, \tau_{\text{share}}\}) \geq \varepsilon > 0$, then every s' with $|s' - s| < \varepsilon$ satisfies $\Gamma(s') = \Gamma(s)$.

Proof. Monotonicity follows because crossing a threshold can only move a score into a weakly higher gate region under $\prec$; within a region Γ is constant. For stability, the open ε -ball about s cannot reach either threshold, hence lies in the same partition cell as s , so Γ is unchanged. $\square$

Lemma 2 (Gate stability implies Bank stability). Under Definition 3, if each s_k stays ε -away from $\{\tau_{\text{suppress}}, \tau_{\text{share}}\}$, then $\{g_k\}$ is unchanged (Lemma 1). For Factor-Bank, the structural recipe in (4) depends only on the qualitative pattern ($g_A = \text{share}$, $g_S = g_R = \text{relearn}$); off-threshold probe noise

SELENE / Factor-LoRA SELECT pipeline

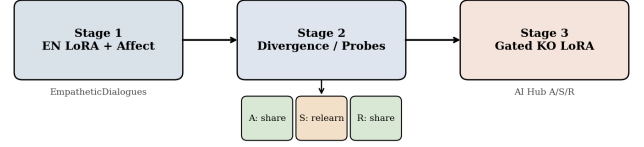


Fig. 1. SELENE overview: English empathy LoRA, factor-wise gate diagnosis, then Factor-Bank KO adaptation (EN-merge + KO LoRA + two-pass A).

Evaluation protocol: Direction I / II

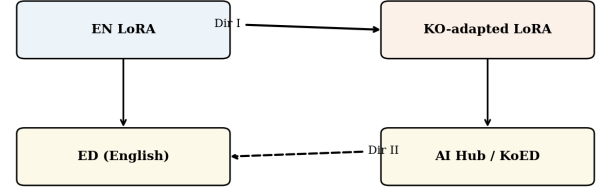


Fig. 2. Evaluation protocol. Direction I measures Korean A/S/R accuracy; Direction II returns to English ED after KO adaptation.

therefore does not flip EN-merge / KO-LoRA / two-pass routing. Gate-conditioned scales (8) are likewise constant within each gate cell.

Proof. Immediate from Lemma 1(ii) and the deterministic map from $\{g_k\}$ to (4)–(8). $\square$

C. Validation Protocol

Propositions 1–4 are *empirical claims*, not existence theorems. We verify them by: (i) KO held-out probes for (1); (ii) cross-factor Acc / Cramér’s V ; (iii) the ordering in Proposition 3; (iv) Factor-Bank vs. Blind / Scratch / soft-share interventions for Proposition 4; (v) stability of $\{g_k\}$ and of the Bank recipe under Lemma 2. Lemmas 1–2 are analytic consequences of the definitions; they do *not* prove that A/C/S/R is the unique empathy factorization, nor that per-factor LoRA tensors are required. Only after (iii)–(iv) hold do we apply Factor-Bank (4) with gate scales (8).

IV. METHOD: SELENE (FACTOR-LoRA SELECT)

SELENE is the system name; Factor-Bank is the Stage 3 realization used in our main results. Unless noted, we freeze Qwen/Qwen3.5–9B—size-matched to KoED open peers (~ 7 –8B) [3]—and train only a *single* KO LoRA bank plus lightweight A/S/R heads. Fig. 1 shows the pipeline; Fig. 2 shows the two evaluation directions; Fig. 3 summarizes the gate rule. We emphasize: factor gates are a *diagnosis*; Factor-Bank is the *implementation* of mixed share/relearn without per-factor LoRA parameters.

A. Theory-Based Factor Definition

Table I lists pre-defined factors and literature anchors. Divergence analysis is used to *validate* these factors, not to

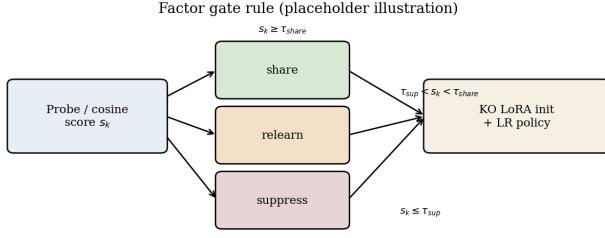


Fig. 3. Factor gate rule mapping probe/similarity scores to share / relearn / suppress (illustrative).

TABLE I
THEORY-ALIGNED EMPATHY FACTORS

Factor	Theory	Operational label
A Affect	Empathic concern [10]	Emotion (ED / AI Hub)
C Cognition	Perspective-taking [10]	Situation text (proxy)
S Strategy	ESC helping skills [2]	Advice / encourage / comfort / agree
R Relation	Display rules + KO norms [3], [11]	AI Hub relation types

TABLE II
MEASURED STAGE 2 FACTOR SCORES AND GATES (QWEN3.5-9B;
 $\tau_{share}=0.55$, $\tau_{suppress}=0.15$).

Factor	Probe Acc	Score s_k	Gate	Notes
A Affect	0.69	0.62	share	probe-heavy mix
S Strategy	0.46	0.46	relearn	EN lacks S labels
R Relation	0.24	0.24	relearn	EN lacks R labels
Culture	—	0.02	suppress	KoED parallel cos.

invent them post hoc from clusters. Davis separates empathic concern (A) from perspective-taking (C) [10]; Rashkin operationalizes emotion + situation in ED [1]; ESC supplies strategy supervision (S) [2]; Matsumoto and KoED motivate relational/cultural display constraints (R) [3], [11].

B. Stage 1: English Empathy LoRA

On EmpatheticDialogues we train a causal LM with LoRA and an affect classification head on pooled prompt states:

$$\mathcal{L}_{EN} = \mathcal{L}_{LM} + \lambda_A \mathcal{L}_{CE}(A). \quad (9)$$

This yields a reusable English empathy adapter M_{EN} while leaving θ intact, consistent with PEFT practice [5].

C. Stage 2: Divergence / Probe Gates

Stage 2 instantiates Definitions 3: with the frozen Stage-1 encoder we (i) fit KO linear probes for A/S/R to obtain s_k in (1), (ii) optionally measure EN-KO / KoED alignment [3], and (iii) apply Γ in (2). When English lacks explicit S/R supervision, we prefer **relearn** unless probe evidence is strong. Fig. 4 and Table II report the measured pattern corresponding to Proposition 3: Affect shares; Strategy/Relation relearn; culture suppresses. These $\{g_k\}$ later drive Factor-Bank routing and gate-conditioned scales (Definition 5), not separate LoRA inits per k .

Factor-wise EN-KO comparison (illustrative placeholders)

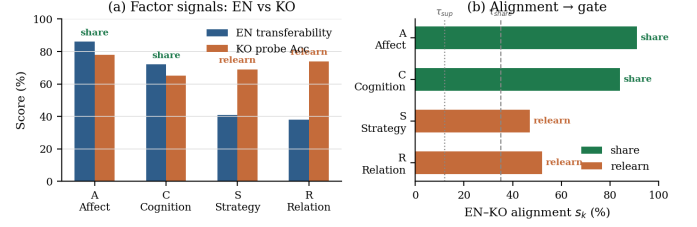


Fig. 4. Factor-wise EN-KO comparison motivating Stage 2 gates. (a) KO probe Acc for A/S/R under the Stage 1 encoder. (b) Scores s_k mapped to share/relearn/suppress (Table II).

Algorithm 1 SELENE Factor-Bank (implemented)

```

Freeze backbone  $Qwen/Qwen3.5-9B$ 
Stage 1: train EN LoRA  $M_{EN}$  + affect head on ED
Stage 2: probe KO; set  $g_k \leftarrow \Gamma(s_k)$  for  $k \in \{A, S, R\}$ 
Stage 3 (Factor-Bank):
  merge  $M_{EN}$  into backbone; init fresh  $M_{KO}$ 
  set  $\lambda_k, \eta_k$  from  $g_k$  (8)
  train with two-pass A:  $f_A(h^{share})$ ,  $f_{S/R}(h^{KO})$ 
return  $M_{EN}$ ,  $M_{KO}$ ,  $\{g_k\}$ 

```

D. Stage 3: Factor-Bank Korean Adaptation

Given mixed gates $g_A=share$, $g_S=g_R=relearn$, we apply Definition 5: merge M_{EN} into the frozen backbone, attach a fresh KO LoRA M_{KO} , and compute affect from the adapter-disabled share path while LM/S/R use the KO path (*two-pass affect*). We train

$$\mathcal{L}_{KO} = \mathcal{L}_{LM} + \lambda_A(g_A)\mathcal{L}_A + \lambda_S(g_S)\mathcal{L}_S + \lambda_R(g_R)\mathcal{L}_R, \quad (10)$$

with gate-conditioned λ_k and head LR (8), plus an early curriculum that further upweights relearn heads [2], [13]. Dir II evaluates the KO LoRA (and heads) on English ED after KO training; we do not claim a separate per-factor composer α in the reported runs. Soft-share / freeze-LoRA ablations keep a single EN-initialized bank and apply gates only through (8)—explaining why soft-share matched Blind until Factor-Bank introduced structural share/relearn routing.

V. EXPERIMENTAL SETUP

A. Datasets and Backbone

Table III summarizes corpora. AI Hub provides full A/S/R labels; ED provides A/C; KoED is held out for cultural evaluation (not training), following [3]. KorEmpatheticDialogues serves as a translate-train control. All main comparisons freeze $Qwen/Qwen3.5-9B$ and match LoRA rank across systems; training curves in Fig. 8 illustrate Stage 1/3 optimization dynamics.

B. Literature Anchor (Mother Paper)

Following KoED’s controlled ED $\leftrightarrow$ KoED comparison [3], Table IV reprints their reported emotion-recognition averages. This anchors Direction I: even strong multilingual models drop on culturally adapted Korean empathy, so “untouched multilingual” is an insufficient upper bound.

TABLE III
DATASET STATISTICS

Corpus	Role	Split size
EmpatheticDialogues	EN Stage 1 / Dir II	17,844 / 2,763 / 2,542
AI Hub Empathetic Dial.	KO Stage 3 / Dir I	25,456 / 3,182
KoED	Cultural eval (held out)	1,360 (test)
KorEmpatheticDialogues	Translate-train baseline	19,531 / 2,769 / 2,547

TABLE IV
KoED EMOTION RECOGNITION ACC (%). REPRINTED AVERAGES FROM [3] (NOT OUR RUNS).

Model	ED (English)		KoED (Korean)	
	w/o j/h	w/ j/h	w/o j/h	w/ j/h
Llama-3.1-8B	34.53	30.74	32.19	28.53
Mistral-7B	39.38	36.40	33.52	31.10
Qwen2-7B	32.73	30.74	25.70	22.57
EXAONE-7.8B	31.41	29.12	30.31	26.18
Claude-3.5	46.17	42.94	41.72	38.31
Average	36.84	33.99	32.69	29.34

TABLE V
SYSTEMS COMPARED (DIRECTION I/II)

System	EN→KO policy	Dir I	Dir II
Untouched KO	KO LoRA from scratch (no EN init)	✓	✓
EN-only	Stage 1 EN LoRA on KO (no KO train)	✓	—
Blind share	Copy EN LoRA; no factor gates	✓	✓
Translate-train (KED)	KoED supervision	✓	✓
MAD-X	Language + task adapters [6]	✓	✓
AdaMergeX [†]	Adaptive adapter merge [7]	—	—
FLARE [†]	EN/target LoRA fusion [8]	—	—
SELENE Bank	EN-merge + KO LoRA + two-pass A	✓	✓

C. Systems

Table V lists systems in a KoED-style checklist. Baselines span KO-scratch, Blind share, translate-train (KorED), and a MAD-X-style dual-adapter control [6]; AdaMergeX/FLARE are cited but not reimplemented under our matched budget [7], [8].

[†]Cited peers; not reimplemented under our matched budget.

D. Direction I — Korean Affective Accuracy

Question: Does Factor-LoRA Bank improve Korean A/S/R over fair Blind/Scratch and MAD-X baselines? Table VI reports full AI Hub valid ($n=3182$, utterance-level S). Blind and Scratch are nearly identical on A Acc; prediction dumps show B16/B32/S16 emotion vectors are *100% identical* on this split—a collapse that Factor-Bank breaks by EN-merge + two-pass affect. F16/F32 gain ≈ 4.5 pp Avg over Blind; our MAD-X control underperforms on Relation despite competitive S.

E. Direction II — Return to English after KO

Question: After KO adaptation, is English emotion recognition preserved? **Metrics:** EmpatheticDialogues valid emotion Acc ($n=2758$); Δ Acc vs. EN-before-KO (65.4%). Table VII: KO-scratch collapses (-34.4 pp). Factor-Bank F16 retains English best (-1.3 pp), beating Blind (-5.9 pp) and F32 (-4.8 pp). Translate-train (KED) retains well (-1.9 pp) but is weak on Dir I A; MAD-X forgets heavily (-26.6 pp).

F. Multi-Seed Stability

Table VIII summarizes seeds $\{123, 456\}$ under the Stage 3 in-training eval protocol (100 batches; R Acc is higher than full-valid). F16 remains ahead of Blind/Scratch on Avg ($\approx +3.4$ pp vs. B16); paired t on Avg yields $t=10.2$ ($p=0.062$, $n=2$ seed pairs). Seed 42 full-valid numbers in Table VI are the primary Dir I claim.

G. Component Ablation

Table IX ablates Factor-Bank components (seed 42, in-training eval Avg). Removing two-pass affect (A2) or EN replay (A5) still yields high Avg, but soft-share alone (A0) and freeze-LoRA (A8) drop toward Blind; LM-only heads (A9) collapse. The structural package—EN-merge + KO relearn LoRA + two-pass A—matters more than any single regularizer.

*Seeds 123/456 in-training Avg; full-valid seed 42 Avg is 79.4 (Table VI).

H. KoED Held-Out Cultural Evaluation

On held-out KoED we map ED 32-way emotion predictions to the 8 shared classes [3]. Table X: Factor-Bank F16 leads 8-class Acc/macro-F1; Scratch collapses; MAD-X fails to transfer culturally; KED is competitive on KoED Acc but weak on AI Hub Dir I.

I. Claims

- **Factor pattern:** Stage 2 assigns Affect=*share* and Strategy/Relation=*relearn* (culture=*suppress*), matching ESC’s fragile strategy axis [2].
- **Dir I:** Factor-Bank gains ≈ 4.5 pp Avg over Blind/Scratch on full valid; Blind=Scratch emotion predictions expose non-selective collapse.
- **Dir II:** F16 forgets only -1.3 pp vs. Scratch -34.4 pp; EN-aware Bank beats Blind retention [9].
- **KoED:** F16 best 8-class Acc/F1 among compared adapters; MAD-X does not close the cultural gap.
- **Ablation:** Soft-share/freeze (gates on loss/LR only) are insufficient; EN-merge + two-pass routing is the operative mechanism.

VI. DISCUSSION

SELENE addresses the cultural empathy gap documented by KoED [3] with a method complementary to language $\times$ task XLT [6]–[8]: factor-wise gate *diagnosis* plus a Factor-Bank that implements mixed share/relearn without per-factor LoRA tensors. Measured Stage 2 scores (Table II) justify selective treatment: strategy and relation carry Korean interactional norms that English empathy adapters do not encode [3], [11], whereas affect transfers more readily [1], [10]. Direction I shows that *structural* routing (EN-merge + KO LoRA + two-pass affect) is necessary—soft-share matched Blind when gates scaled only losses/LRs on one shared bank, and Blind/Scratch emotion predictions are identical on AI Hub valid. Direction II and KoED jointly argue that Bank is not a Dir I-only trick: F16 best preserves English Acc and leads held-out cultural emotion recognition. This view is consistent

TABLE VI
DIRECTION I: FULL AI HUB VALID ($n=3182$), QWEN3.5-9B BF16 LoRA. AVG = MEAN(A ACC, S F1, R ACC).

System	A Acc $\uparrow$	S F1 $\uparrow$	R Acc $\uparrow$	Avg $\uparrow$
KO-scratch (S16)	80.9	62.5	80.5	74.6
Blind share (B16)	80.9	62.7	81.0	74.9
Blind +MLP (B32)	80.9	62.8	81.2	75.0
MAD-X (lang+task)	78.4	64.3	67.4	70.1
SELENE Bank (F16)	87.7	64.2	86.3	79.4
SELENE Bank +MLP (F32)	87.0	64.0	87.4	79.5

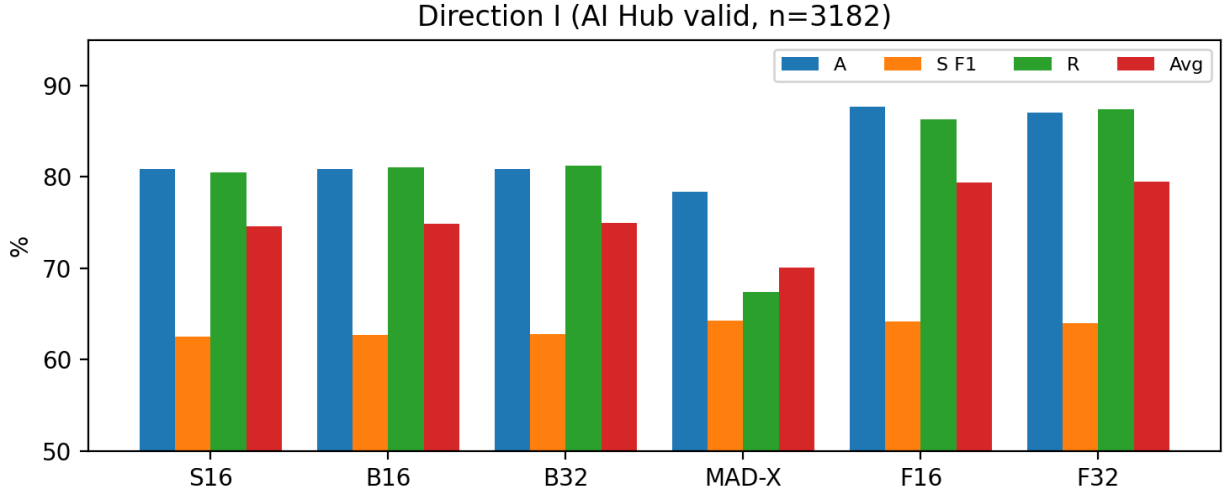


Fig. 5. Direction I A/S/R Acc and Avg on AI Hub valid (Table VI).

TABLE VII
DIRECTION II: RETURN-TO-EN ON EMPATHETICDIALOGUES VALID ($n=2758$).

System (adapter on ED)	Acc $\uparrow$	Δ Acc
EN before KO	65.4	0.0
After KO-scratch (S16)	31.0	-34.4
After Blind share (B16)	59.4	-5.9
After Blind +MLP (B32)	58.9	-6.5
After MAD-X	40.1	-26.6
After KED (translate-train)	64.8	-1.9
After SELENE Bank (F16)	64.1	-1.3
After SELENE Bank (F32)	60.6	-4.8

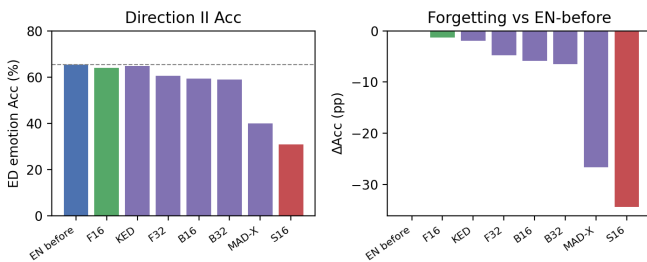


Fig. 6. Direction II return-to-EN Acc and Δ Acc (Table VII).

with cultural-dominance findings in LLMs [17], multidimensional empathy theory [10], [13], and less-forgetting multilingual fine-tuning [9], [22].

TABLE VIII
MULTI-SEED DIR I (IN-TRAINING EVAL, MEAN $\pm$ STD; SEEDS 123/456).

System	A	S	R	Avg
F16 (Bank)	89.0 $\pm$ 1.4	66.2 $\pm$ 1.6	97.0 $\pm$ 0.0	84.1$\pm$1.0
B16 (Blind)	77.5 $\pm$ 0.0	68.0 $\pm$ 0.8	96.5 $\pm$ 0.7	80.7 $\pm$ 0.5
S16 (Scratch)	77.5 $\pm$ 0.0	67.5 $\pm$ 0.1	97.2 $\pm$ 0.4	80.8 $\pm$ 0.1

TABLE IX
COMPONENT ABLATION OF SELENE BANK (SEED 42; IN-TRAINING DIR I AVG).

ID	Configuration	Avg $\uparrow$
F16	Full Factor-Bank	83.4–84.8*
A0	Soft-share (v3-style)	79.7
A2	w/o two-pass affect	86.0
A3	w/o gate-conditioned losses	83.9
A4	w/o select curriculum	82.9
A5	w/o EN replay	85.2
A6	w/o LoRA anchor	82.8
A8	Freeze EN LoRA (heads only)	77.0
A9	LM only (no A/S/R heads)	34.0

a) *Limitations.*: We do *not* train separate LoRA modules $\{M_k\}_{k \in \mathcal{K}}$; Factor-Bank uses one KO adapter with two-pass

TABLE X
KoED HELD-OUT 8-CLASS SHARED EMOTION ACC / MACRO-F1 (%).

System	Acc↑	macro-F1↑
After Scratch (S16)	68.2	38.4
After Blind (B16)	76.5	62.7
After Blind (B32)	77.9	63.9
After MAD-X	18.4	12.7
After KED	76.9	72.6
After SELENE Bank (F16)	84.1	74.0
After SELENE Bank (F32)	76.6	65.8

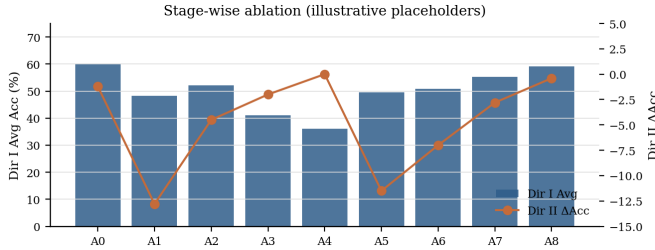


Fig. 7. Component ablation overview (Table IX).

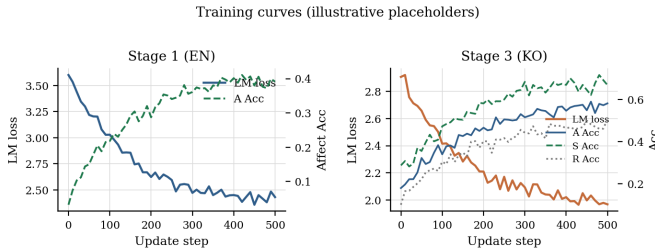


Fig. 8. Stage 1 (EN) and Stage 3 (KO) training curves.

routing and gate-conditioned λ_k/η_k . Fully factor-separated LoRA banks (literal $M_{KO}=\{M_k\}$) remain future work and would be needed to claim parameter-level factor gating. AdaMergeX/FLARE are not reimplemented under matched compute; Llama generalization was skipped pending Stage 1. Multi-seed tables use the in-training eval protocol; primary Dir I claims use full valid ($n=3182$). AI Hub licensing constrains redistribution of raw dialogues. KoED human cultural-appropriateness (CA) scoring remains future work.

VII. CONCLUSION

We presented **SELENE**, a selective empathy LoRA system for English–Korean affective transfer on frozen Qwen3.5-9B. Stage 2 gates diagnose Affect as shareable and Strategy/Relation as releasable; Factor-Bank implements that *mixed* diagnosis with EN-merge, one KO LoRA, two-pass affect, and gate-conditioned losses/LRs—not per-factor LoRA tensors. On full AI Hub valid, Bank improves Dir I Avg by ≈ 4.5 pp over Blind/Scratch, retains English emotion Acc within 1.3 pp (vs. Scratch -34.4 pp), and leads KoED 8-class Acc/F1. Empathy

does not transfer as a single skill [2], [3], [10]: factor-aware routing and optimization—not blind adapter copy—make EN→KO affective transfer both accurate and stable.

ACKNOWLEDGMENTS

The authors thank the maintainers of EmpatheticDialogues, KoED, and AI Hub for releasing resources that enable culturally grounded empathy research.

REFERENCES

- [1] H. Rashkin, E. M. Smith, M. Li, and Y.-L. Boureau, “Towards empathetic open-domain conversation models: A new benchmark and dataset,” in *Proc. ACL*, 2019, pp. 5370–5381.
- [2] S. Liu *et al.*, “Towards emotional support dialog systems,” in *Proc. ACL*, 2021, pp. 3469–3483.
- [3] W. Lee, Y. Sim, H. Kim, and H. Kim, “Multilingual, not multicultural: Uncovering the cultural empathy gap in LLMs through a comparative empathetic dialogue benchmark,” in *Proc. IJCNLP-AACL*, 2025, pp. 791–809.
- [4] S. Sabour *et al.*, “EmoBench: Evaluating the emotional intelligence of large language models,” in *Proc. ACL*, 2024.
- [5] E. J. Hu *et al.*, “LoRA: Low-rank adaptation of large language models,” in *Proc. ICLR*, 2022.
- [6] J. Pfeiffer, I. Vulić, I. Gurevych, and S. Ruder, “MAD-X: An adapter-based framework for multi-task cross-lingual transfer,” in *Proc. EMNLP*, 2020, pp. 7654–7673.
- [7] Y. Zhao *et al.*, “AdaMergeX: Cross-lingual transfer with large language models via adaptive adapter merging,” in *Proc. NAACL*, 2025.
- [8] P. N. Borchert *et al.*, “Language fusion for parameter-efficient cross-lingual transfer,” in *Proc. ACL*, 2025.
- [9] Y. Mao, Y. Liang, N. Duan, H. Wang, K. Wang, L. Chen, and Y. Gao, “Less-forgetting multi-lingual fine-tuning,” in *Proc. NeurIPS*, 2022.
- [10] M. H. Davis, “Measuring individual differences in empathy: Evidence for a multidimensional approach,” *J. Pers. Soc. Psychol.*, vol. 44, no. 1, pp. 113–126, 1983.
- [11] D. Matsumoto *et al.*, “Culture, emotion regulation, and adjustment,” *J. Pers. Soc. Psychol.*, vol. 94, no. 6, pp. 925–937, 2008.
- [12] M. M. Amin, E. Cambria, and B. W. Schuller, “A wide evaluation of ChatGPT on affective computing tasks,” *IEEE Trans. Affect. Comput.*, 2024.
- [13] A. Sharma, A. Miner, D. Atkins, and T. Althoff, “A computational approach to understanding empathy expressed in text-based mental health support,” in *Proc. EMNLP*, 2020, pp. 5263–5276.
- [14] Z. Lin *et al.*, “MoEL: Mixture of empathetic listeners,” in *Proc. EMNLP-IJCNLP*, 2019, pp. 133–143.
- [15] N. Majumder *et al.*, “MIME: Mimicking emotions for empathetic response generation,” in *Proc. EMNLP*, 2020, pp. 8968–8979.
- [16] S. Sabour, C. Zheng, and M. Huang, “CEM: Commonsense-aware empathetic response generation,” in *Proc. AAAI*, 2022, pp. 11229–11237.
- [17] W. Wang *et al.*, “Not all countries celebrate Thanksgiving: On the cultural dominance in large language models,” in *Proc. ACL*, 2024, pp. 6349–6384.
- [18] K. Ahuja *et al.*, “MEGA: Multilingual evaluation of generative AI,” in *Proc. EMNLP*, 2023, pp. 4232–4267.
- [19] E. M. Ponti *et al.*, “XCOPA: A multilingual dataset for causal commonsense reasoning,” in *Proc. EMNLP*, 2020, pp. 2362–2376.
- [20] A. Conneau *et al.*, “Unsupervised cross-lingual representation learning at scale,” in *Proc. ACL*, 2020, pp. 8440–8451.
- [21] N. Houlsby *et al.*, “Parameter-efficient transfer learning for NLP,” in *Proc. ICML*, 2019, pp. 2790–2799.
- [22] J. Kirkpatrick *et al.*, “Overcoming catastrophic forgetting in neural networks,” *Proc. Natl. Acad. Sci. U.S.A.*, vol. 114, no. 13, pp. 3521–3526, 2017.

Seohyeon Yoo Seohyeon Yoo received the Ph.D. degree and is with the Department of Software Engineering, Jeonbuk National University, Jeonju, Republic of Korea.

Jaehyuk Cho Jaehyuk Cho received the Ph.D. degree and is with the Department of Software Engineering, Jeonbuk National University, Jeonju, Republic of Korea (e-mail: chojh@jbnu.ac.kr). He is the corresponding author.